

# CHRISTOPHER MIOTTO

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**Summary:** Ph.D. Researcher specializing in domain-specific architectures and hardware/software co-design.

## EDUCATION

### The University of Michigan

Doctor of Philosophy

Major in Computer Science and Engineering

Ann Arbor, Michigan

Expected May 2031

### Purdue University

Master of Science

Major in Electrical and Computer Engineering

West Lafayette, Indiana

May 2026

GPA: 4.0/4.0

Bachelor of Science

Major in Computer Engineering; Minor in Business Economics

May 2025

GPA: 3.98/4.0

## WORK EXPERIENCE

### Purdue University

Graduate Teaching Assistant

West Lafayette, Indiana

Aug 2025 – May 2026

- Directed 4 lab sections, supervised Undergraduate TAs, and guided 100+ students to deliver dual-core RISC-V processors.
- Conducted 12 weekly hours of debugging sessions, managing Xilinx FPGA boards to validate designs via physical synthesis.

AI Hardware Team Researcher – Purdue System-On-Chip Extension Technologies (SoC-ET)

Jan 2025 - May 2026

- Designed a tournament branch predictor in SystemVerilog, reducing misprediction penalties by ~60% for an O3 system.
- Developed a custom vector core microarchitecture optimized for RISC-V vector extensions targeting AI acceleration.

### Radiance Technologies

ASIC Design Intern

Huntsville, Alabama

May 2025 - Aug 2025

- Verified 10+ synthesizable RTL modules in SystemVerilog, resolving synthesis-simulation mismatches in Cadence Xcelium.
- Scripted Genus synthesis automation in Tcl, reducing manual configuration time and optimizing gate-level netlist generation.

## PROJECTS

### GPGPU-Sim Microarchitecture Enhancements: Register File Caching & Warp Scheduling

Winter 2026

- Engineered a Register File Cache via *opndcoll\_rfu\_t* C++ edits to bypass arbiter MRF reads.
- Dynamically reallocated per-warp RFC capacity, driving a 62% Rodinia MRF read reduction.
- Altered *two\_level\_active\_scheduler* with a grace-period policy, cutting warp demotions by 7.8%.

### gem5 Multiprocessor Ruby Coherence: RegionScout Implementation

Winter 2026

- Engineered C++ NSRT and CRH structures to extend the gem5 Ruby *MOESI\_hammer* coherence protocol.
- Integrated structure updates into cache state transitions and modified directory actions to bypass region exclusive snoops.
- Suppressed Splash-4 memory traffic, cutting control broadcasts by 46% and total interconnect traffic by 37%.

### gem5 Microarchitecture O3 Enhancements: Waiting Instruction Buffer & Victim Cache

Fall 2025

- Engineered an O3 Waiting Instruction Buffer via *iem.cc* and scoreboard modifications to divert L2 miss dependents.
- Exposed independent work to resolve pointer-chasing pipeline stalls, achieving a 1.8x speedup on the MCF benchmark.
- Altered *allocOnFill* logic in *base.bb* for a Strict Exclusive victim cache, reducing NAB miss rates by 65%.

### SystemVerilog RTL Design: Dual-Core 5-Stage-Pipelined RISC-V Processor with Cache Coherence

Fall 2024

- Implemented split caches to reduce memory access latency and contention alongside MSI Protocol for cache coherence.
- Demonstrated a 3.45x speedup from caching and a 5.0x speedup from multicore over single-cycle baseline processor.

## SKILLS

**Hardware Description & Assembly:** SystemVerilog, RISC-V Assembly

**Software & Scripting:** C/C++, Python, CUDA, Tcl/Shell Scripting

**Simulation & EDA Tools:** gem5, GPGPU-Sim, Cadence (Genus, Xcelium), Siemens (QuestaSim)

**Security Clearance:** Secret (Active), U.S. Department of Defense, Adjudicated June 2025